Goldman Sachs Research

Equity Research

12 October 2026 | 5:00AM EDT

GLOBAL TECHNOLOGY: SEMICONDUCTORS - MEMORY

High-performance NAND could help crumble the AI "memory wall" while driving meaningful CapEx savings

James Schneider, Ph.D.
+1(212)357-2929 | jim.schneider@gs.com
Goldman Sachs & Co. LLC

Khalil Fenina
+1(212)357-6392 | khalil.fenina@gs.com
Goldman Sachs & Co. LLC

Giuni Lee
+65-6889-1025 | giuni.lee@gs.com
Goldman Sachs (Singapore) Pte

Shuhei Nakamura
+81(3)4587-9932 | shuhei.nakamura@gs.com
Goldman Sachs Japan Co., Ltd.

Allen Chang
+852-2978-2930 | allen.k.chang@gs.com
Goldman Sachs (Asia) L.L.C.

Daiki Takayama
+81(3)4587-9870 | daiki.takayama@gs.com
Goldman Sachs Japan Co., Ltd.

Marshall Wong
+1(646)446-0222 | marshall.wong@gs.com
Goldman Sachs & Co. LLC

Anmol Makkar
+1(212)357-1366 | anmol.makkar@gs.com
Goldman Sachs & Co. LLC

Luya You
+1(212)902-5297 | luya.you@gs.com
Goldman Sachs & Co. LLC

Liana Wadhwani
+1(212)357-3204 | liana.wadhwani@gs.com
Goldman Sachs & Co. LLC

Key stock takeaways: We view emerging high-performance NAND-based solutions - including High-Bandwidth Flash (HBF) - as promising tools to address a worsening memory constraint in AI inference. We see a potential TAM of ~$40 - $150B at a GPU adoption rate of 15%-50% by 2030 given the rapid proliferation of read-heavy and memory-intensive AI inference workloads - despite potential challenges to adoption. We believe that HBF memory can meaningfully reduce the overall cost of inference throughput by increasing GPU utilization, resulting a ~30% lower overall cost of inference throughput - translating to as much as several hundreds of billions of hyperscaler CapEx savings.

In a bull-case scenario that assumes rapid commercial adoption of HBF on 50% of XPUs by late 2028, we see HBF bits accounting for ~4% of global NAND bit shipments - but ~9% of NAND wafer capacity given its die size penalties - similar to the HBM/DRAM dynamic. We would point out that these NAND-based solutions remain in the early development stages at a system level, and that a wide range of outcomes exists that could drive a significantly greater or smaller impact to the overall NAND supply/demand market dynamic. Stepping back, we continue to believe that limited supply additions will prolong tight NAND supply/demand in the near term - and view HBF as potentially extending the duration of the current cycle. While HBF is the technology approach that has received the most attention in the technical literature, we would also highlight XL-Flash (from Kioxia) and zNAND-O (from Samsung) as alternative approaches to achieving similar objectives.

We continue to favor SanDisk (SNDK, Buy) and SK Hynix (000660.KS) given their technology leadership in HBF, and strong core business with improving product mix. We also see Samsung Electronics (005930.KS, Buy) as well positioned given its strong NAND portfolio; we see Micron (MU, Neutral) as less of a direct beneficiary given its primary focus on DRAM and lower exposure to NAND. Although we recognize challenges to widespread HBF adoption, we see promise in the technology which could drive meaningfully higher cost/performance at the system level, particularly in memory-intensive inference workloads.

Goldman Sachs does and seeks to do business with companies covered in its research reports. As a result, investors should be aware that the firm may have a conflict of interest that could affect the objectivity of this report. Investors should consider this report as only a single factor in making their investment decision. For Reg AC certification and other important disclosures, see the Disclosure Appendix, or go to www.gs.com/research/hedge.html. Analysts employed by non-US affiliates are not registered/qualified as research analysts with FINRA in the U.S.

Executive Summary

On balance, we are constructive on HBF technology, and it as a promising potential solution to the intensifying memory constraint in AI inference. Our bullish view on the technology's potential is based on the following:

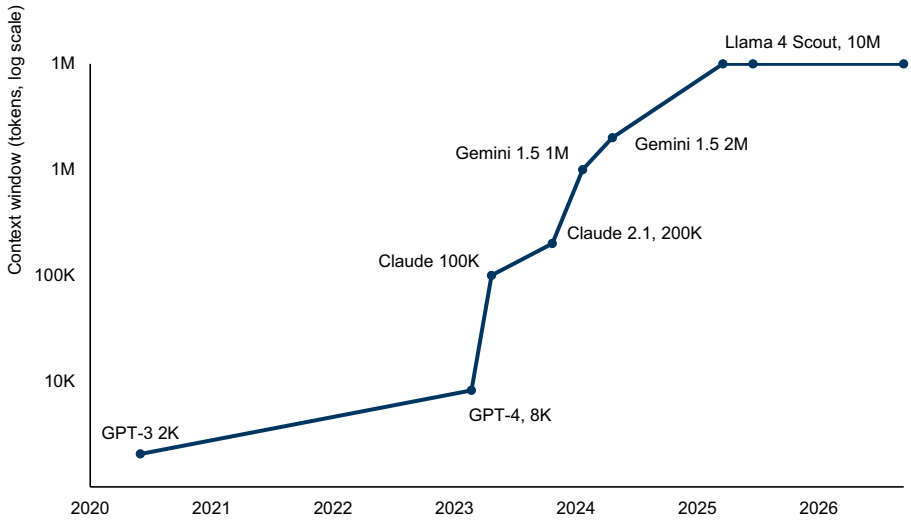
1. Proliferation of AI inference workloads will lead to structurally higher demand for high-bandwidth, near-compute storage as context windows lengthen.
2. Highly cost-competitive near-compute memory capacity at comparable sequential read bandwidth to HBM could result in attractive system-level economics and accelerate adoption of HBF among hyperscalers.
3. NAND's high density of NAND, coupled with HBF's higher resulting GPU utilization, will allow the technology to command a meaningful price premium while simultaneously lowering system-level costs for hyperscalers and AI labs.

However, key challenges to adoption remain, namely latency concerns (particularly for random read / write operations), endurance and thermal considerations, and manufacturing resource intensity. Despite these challenges, we see a significant commercial opportunity in HBF and view the economics of its adoption as attractive - even at a significant price premium relative to traditional NAND.

Background - Why is there a need for High-Bandwidth Flash?

The shift to inference requires much near-compute storage capacity, and HBF's similar sequential read bandwidth vs. HBM positions it well as a potential solution to the memory constraint given NAND's high density. As more AI workflows shift away from compute-heavy training and toward memory-intensive inference, context windows are lengthening exponentially and scaling toward 10 mn+ tokens from a ~1.5 mn baseline (see Exhibit 1). The KV (key value) cache required for these more memory-intensive workloads is growing with larger context windows, broadly shifting the memory bottleneck from purely bandwidth-related to a combined bandwidth-capacity constraint. Although we believe DRAM-based High Bandwidth Memory (HBM) - and its subsequent iterations - is likely to lead in bandwidth and performance, the technology is significantly capacity-constrained (current architectures typically include ~192 GB per GPU) given current DRAM supply constraints. AI infrastructure operators are thus deploying additional GPU units to drive incremental compute-adjacent memory capacity, resulting in meaningfully underutilized compute capacity. Through a strategic collaboration, SanDisk and SK Hynix have introduced High Bandwidth Flash (HBF) as a proposed solution to this near-compute capacity constraint.

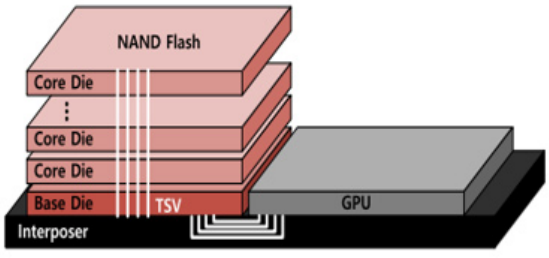
Exhibit 1: Chart showing context window growth over time for common LLMs



Source: Hidekazu-Konishi, data compiled by Goldman Sachs Global Investment Research

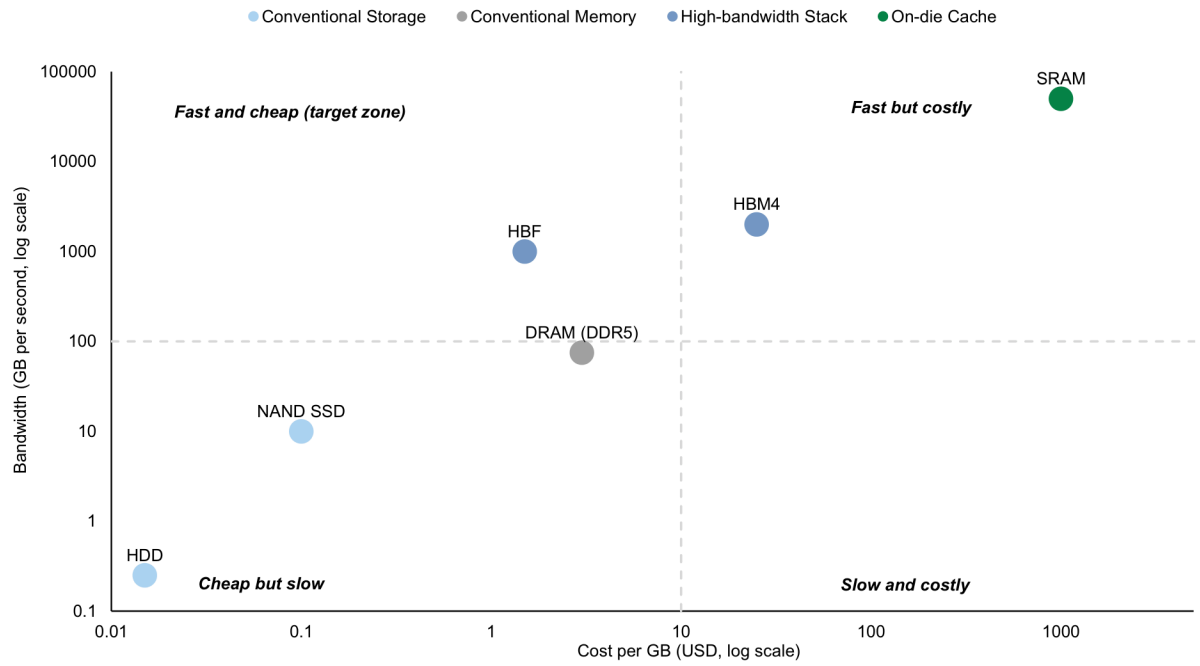
We expect HBF pricing to be significantly below DRAM-based HBM4 contract pricing (on a per GB basis) once commercial samples begin shipping in CY27 (see Exhibit 3), despite the product's pricing currently being undisclosed. High Bandwidth Flash (HBF) has been introduced an open-standard, near-compute memory tier that stacks high-density 3D NAND flash into the AI accelerator package via through-silicon vias (TSVs) and UCIe chiplet interconnects (see Exhibit 2). It establishes an intermediate tier in the memory hierarchy, offering a bandwidth leap over standard NAND SSDs and offers near-HBM bandwidth on a sequential read basis. Governed under the Open Compute Project (OCP) which includes SanDisk and SK Hynix, as well as ASIC customer hyperscalers (namely Alphabet and Meta), HBF delivers up to 512 GB per stack (vs. 16-32 GB per HBM stack) at sequential read bandwidths up to ~3.0 TB/s (see Exhibit 5), per the technical literature.

Exhibit 2: Cross-sectional architecture of High Bandwidth Flash (HBF)



Source: SK Hynix

Exhibit 3: Memory hierarchy Diagram



Source: Goldman Sachs Global Investment Research

Advantages of HBF - Cost effectiveness and other benefits

We believe HBF can lower the overall cost of equivalent inference token output by ~30% or more, and lead to significant reductions in hyperscaler capital expenditures through combined hybrid architectures with HBM and increased GPU utilization. Given NAND's thermal and write endurance constraints, we believe HBF stacks are likely to utilize SLC-based (single level cell) NAND to mitigate these risks - and given the higher costs and die size penalties will be priced far above MLC - similar to the relationship between HBM and conventional DRAM. HBF's interconnect specification will likely also result in significantly lower output per wafer (given the larger physical size of an HBF chip relative to an equivalent capacity standard NAND chip; we assume HBF requires ~3.5X more die/wafer area vs. traditional NAND). Even at a meaningful premium, we see significant room for HBF to significantly lower inference output costs, even at ~1mn token context length - with the benefit growing roughly linearly as sequences scale higher (see Exhibit 4).

Exhibit 4: Illustrative per 1mn token cost comparison HBM only vs. H3 GPU memory architecture and HBF price per GB scenario analysis

On-Chip Memory Configuration	HBM Only	HBM-HBF Hybrid
GPU Model	Nvidia B200	Nvidia B200
HBM Capacity (GB) per GPU	192	192
HBF Capacity (GB) per GPU	0	4,096
Number of GPUs	**8**	**4**
Total System HBM Capacity	1,536	768
Total System HBF Capacity	0	16,384
HBM Cost per GB ($)	15.00	15.00
HBF Cost per GB ($)	1.75	1.75
HBM Cost per GPU ($)	2,880	2,880
HBF Cost per GPU ($)	0	7,168
GPU Cost (incl. packaging, peripherals, but excl. memory)	7,620	7,620
Aggregate System Cost (GPUs + Memory)	84,000	70,672
Aggregate System Price (GPUs + Memory) (70% GM)	**280,000**	**235,573**
Illustrative Inference Throughput in TPS at a 1mn token batch size (approximate)	10,000	12,500
Cost per 1K inference output tokens per second	**$28.00**	**$18.85**
Cost Efficiency Gain		-32.69%

Potential Cost Efficiency Gain for H3 at 4 GPUs	
HBF price per GB	
$ 0.25	-56.10%
$ 0.50	-52.20%
$ 0.75	-48.30%
$ 1.00	-44.40%
$ 1.25	-40.50%
$ 1.50	-36.59%
$ 1.75	-32.69%
$ 2.00	-28.79%
$ 2.25	-24.89%
$ 2.50	-20.99%

Source: Company data, TrendForce, Goldman Sachs Global Investment Research

HBF could achieve meaningful cost advantages over HBM-only systems with higher memory density, comparable sequential read bandwidth, and higher GPU utilization. We highlight key technical points including:

- **Higher memory density:** At ~512 GB per stack and ~4 TB per GPU (in SK Hynix's proposed H3 architecture), HBF offers ~16X higher memory capacity vs. HBM. This density allows systems to process 2.6X larger batch sizes at 1M tokens and up to 18.8X larger batches at 10M tokens compared to pure-HBM setups - resulting in serving throughput advantages widen alongside sequence length. This advantage is estimated at ~1.25X at 1 mn tokens and ~6.14X at 10 mn tokens, according to research published by SK Hynix.
- **Mitigating the bandwidth gap between DRAM and NAND:** Although NAND lags DRAM in peak bandwidth, research from SK Hynix and SanDisk has shown that sequential read bandwidth for HBF significantly narrows the gap with HBM and achieves up to ~3 TB/s (see Exhibit 5). Also, in most inference-heavy workloads, the number of write operations is quite limited, reducing the significance of random read or write bandwidth of memory.
- **~2.7X energy efficiency gain at scale:** Although an individual HBF stack consumes 2X to 4X more power in absolute terms (80-160W vs. 40W), its ability to process significantly larger inference batches may yield up to ~2.7X higher throughput-per-watt, with efficiency gains widening as context lengths scale, according to research supported by SK Hynix.
- **Potential for XPU cost savings via higher utilization:** By eliminating memory-capacity bottlenecks and maximizing GPU core utilization, 4 HBF-equipped XPU accelerators can potentially match the aggregate serving throughput of an 8-XPU pure-HBM system (at 192 GB / 4 TB of HBM / HBF respectively per GPU,

according to SanDisk).

Exhibit 5: Comparison between HBF and HBM 4

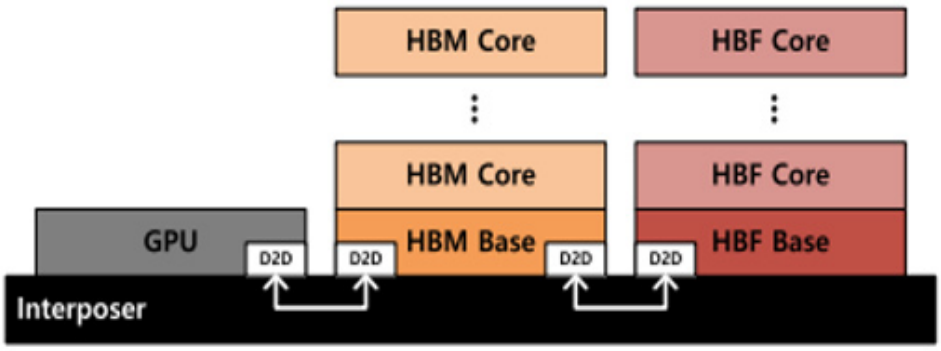
Comparison between HBF and HBM 4		
Category	HBM 4	HBF
Capacity (GB)	36	512
Price ($)	~15.28	N/A
Power Consumption (Watt)	HBF is ~2-4x HBM	
Throughput per power	HBF 1.25x higher for 1M tokens; 6.14x higher for 10M *	
Sequential Read Bandwidth	~2.0-3.3 TB/s	~0.4-3 TB/s
Latency	~20ns	~10 µs (10000ns)
Read granularity (bytes per read)	32	4096
Write Endurance	High	Low
Excels in	Write dominant workload	Read dominant workload

Source: Silicon analyst, Hotchips conference, Goldman Sachs Global Investment Research

HBF's benefits are enhanced in hybrid architectures that help mitigate the technology's key drawbacks. Since NAND flash access latency may prove sub-optimal for real-time token generation in certain situations, SK Hynix has proposed its own "H^3" architecture, which consists of a hybrid HBM-HBF configuration. The company claims this helps address NAND's write / random read bandwidth shortcomings by unifying both memory types into a complementary, two-tier memory hierarchy. We note the combined architecture could potentially offer:

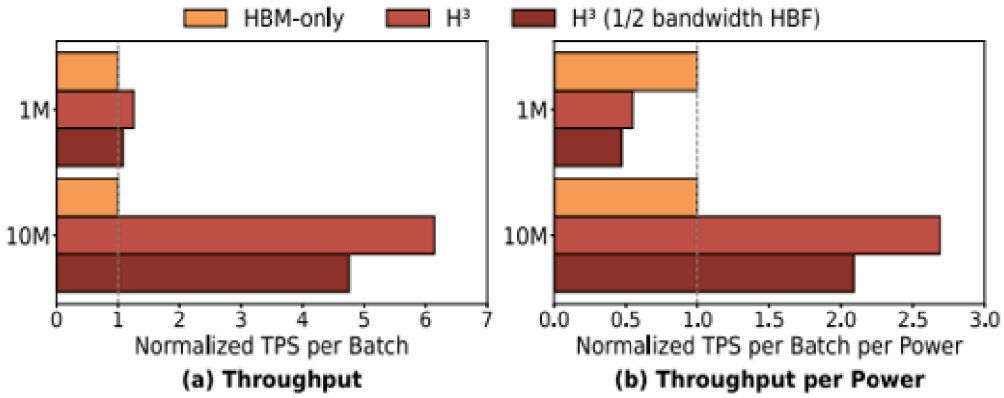
- **Zero shoreline penalty:** HBF stacks are daisy-chained directly through the HBM base die, expanding on-package capacity without occupying the GPU's limited physical edge connections (see Exhibit 6)
- **Optimized workload partitioning:** Low-cost HBF stores massive, static data structures (model weights and pre-computed KV caches), reserving valuable HBM capacity exclusively for latency-critical live decoding.
- **Commercial and operational payoff:** Offloading static memory expands concurrent batch capacity, enabling up to 2.7x higher throughput per watt (see Exhibit 7). This reduces overall GPU cluster counts and facility power draw, lowering total cost of ownership (TCO) and per-query serving costs.

Exhibit 6: Side cross sectional view of H3



Source: SK Hynix

Exhibit 7: Throughput & throughput per power comparison of HBM-only vs H3



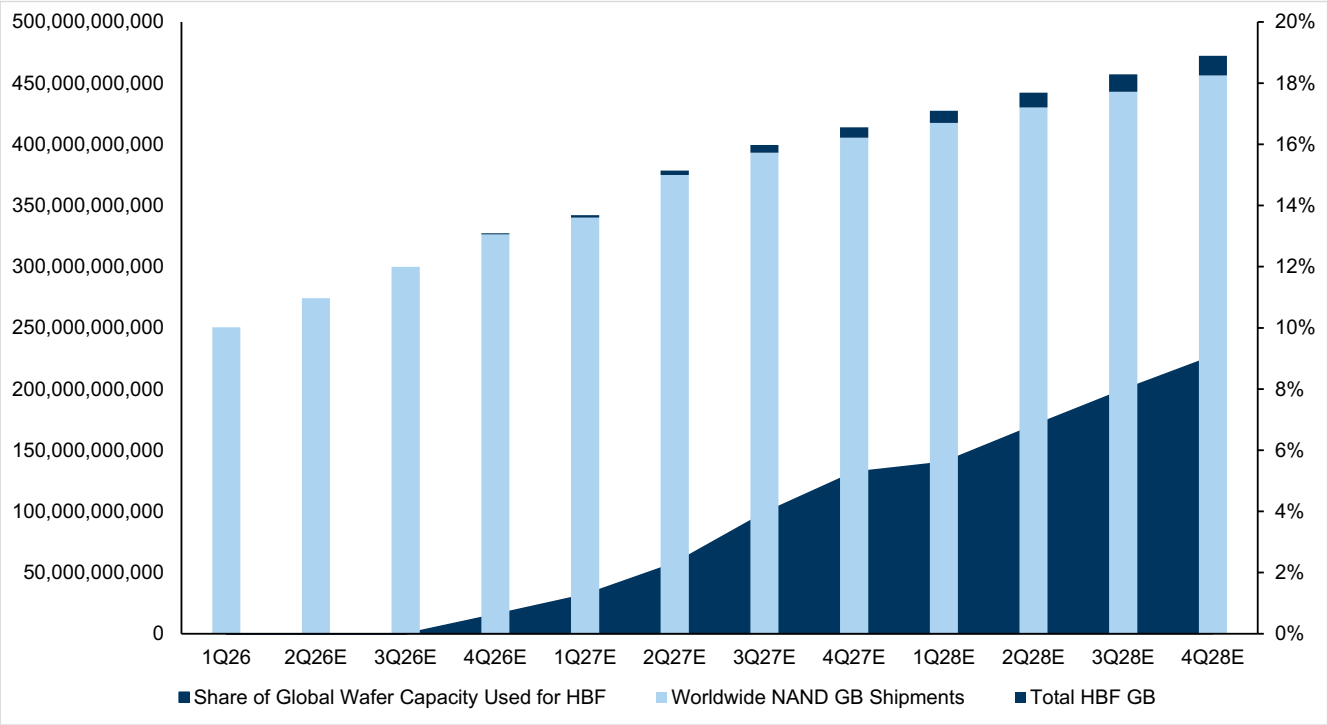
Source: SK Hynix

HBF forecast model and potential commercial opportunity

We see a significant potential commercial opportunity in HBF and believe it could account for up to ~4% of global NAND bit shipments by late 2028 in a bull case that assumes a rapid and elevated adoption rate of 50% by 4Q28, which could equate to an HBF TAM of ~$150 bn. In a more measured scenario that assumes a penetration rate of ~15% of GPU units by late 2030 and a price per GB of HBF of $1.75 (which we view as conservative), we see an annualized TAM for HBF of ~$40 bn.

We see HBF bits occupying up to ~9% of global NAND wafer capacity in a bull-case adoption scenario given its SLC structure and larger die size. Relative to traditional NAND, we assume a 3.5X higher wafer space usage per GB equivalent for HBF (roughly similar to the DRAM/HBM ratio) and thus see it accounting for a larger share of global NAND wafer capacity relative to its share of total NAND bits. Assuming a bull case scenario of a rapid ramp in HBF's deployment to 50% of GPU units by late 2028, and based on current industry forecasts for global NAND bit output and wafer capacity, we see HBF accounting for ~4% of global NAND bits while simultaneously consuming over 9% of global NAND wafer capacity (see Exhibit 8).

Exhibit 8: HBF share of global NAND GB and global NAND wafer capacity



Source: TrendForce, Goldman Sachs Global Investment Research

Exhibit 9: Global HBF forecast model

	1Q26	2Q26E	3Q26E	4Q26E	1Q27E	2Q27E	3Q27E	4Q27E	1Q28E	2Q28E	3Q28E	4Q28E
NVDA	1,592	1,866	2,067	2,265	2,525	2,393	3,093	3,345	3,402	3,447	3,575	3,588
AMD	127	135	146	244	266	298	343	377	358	362	365	376
AVGO	1,254	1,379	1,614	1,904	1,828	2,193	2,522	2,951	2,951	3,305	3,438	3,541
MRVL	110	118	148	195	152	178	208	239	206	237	278	333
Total (k Units)	3,082	3,499	3,975	4,608	4,771	5,062	6,166	6,912	6,917	7,351	7,656	7,838
% w/ HBF	0.00%	0.00%	0.00%	5.00%	10.00%	17.50%	25.00%	30.00%	35.00%	40.00%	45.00%	50.00%
k Units w/ HBF	0	0	0	230	477	886	1,541	2,074	2,421	2,940	3,445	3,919
HBF GB per Unit	4,096	4,096	4,096	4,096	4,096	4,096	4,096	4,096	4,096	4,096	4,096	4,096
Total HBF GB	0	0	0	943,769,934	1,954,354,278	3,628,252,884	6,313,882,638	8,493,715,308	9,916,832,398	12,044,213,194	14,111,129,075	16,051,941,696
Global NAND GB Shipped	250,502,593,784	274,274,827,172	299,837,605,810	326,345,845,374	340,103,507,470	374,978,692,941	393,082,944,774	405,412,980,472	417,575,369,886	430,102,630,983	443,005,709,912	456,295,881,210
QoQ Growth (%)		9.32%	8.84%	4.22%	10.25%	4.83%	3.14%	3.00%	3.00%	3.00%	3.00%	
% of WW NAND GB as HBF	0.00%	0.00%	0.00%	0.29%	0.57%	0.97%	1.61%	2.10%	2.37%	2.80%	3.19%	3.52%
Estimated SanDisk GB Shipped	80,147,536,031	80,147,536,031	80,147,536,031	80,147,536,031	1.01133E+11	1.01133E+11	1.01133E+11	1.01133E+11	1.11246E+11	1.11246E+11	1.11246E+11	1.11246E+11
% of SanDisk GB Shipped (illustrative, 100% market share)	0.00%	0.00%	0.00%	1.18%	1.93%	3.59%	6.24%	8.40%	8.91%	10.83%	12.68%	14.43%
Global Wafer Starts in NAND	4,011,000	4,026,000	4,092,000	3,987,000	4,071,000	4,191,000	4,263,000	4,293,000	4,722,300	4,816,746	4,913,081	5,011,343
Max Trad. NAND GB per 12in. Wafer	131,072	131,072	131,072	131,072	131,072	131,072	131,072	131,072	131,072	131,072	131,072	131,072
HBF to Trad. NAND Wafer Space Ratio	3.5x	3.5x	3.5x	3.5x	3.5x	3.5x	3.5x	3.5x	3.5x	3.5x	3.5x	3.5x
Max HBF GB per 12in. Wafer	37,449	37,449	37,449	37,449	37,449	37,449	37,449	37,449	37,449	37,449	37,449	37,449
Wafers Needed for HBF	-	-	-	25,201	52,187	96,885	168,599	226,807	264,808	321,615	376,808	428,633
Share of Global Wafer Capacity Used for HBF	0.00%	0.00%	0.00%	0.63%	1.28%	2.31%	3.95%	5.28%	5.61%	6.68%	7.67%	8.55%

Source: TrendForce, Goldman Sachs Global Investment Research

Challenges to HBF adoption

Despite the promise of High-Bandwidth Flash, we note several key challenges to adoption.

- **Bandwidth constraints and latency concerns:** According to SanDisk, SK Hynix, and other studies, HBF offers comparable sequential read bandwidth to HBM4, although

we note that (1) HBM4E is currently in the sampling phase and delivers at least ~50% higher bandwidth; (2) although inference workloads are read-dominant overall, any write operations significantly reduces the effective bandwidth of HBF given its NAND-based design; (3) HBF's structurally lower write performance and endurance relative to DRAM-based HBM results in its practical benefits being highly dependent on its read performance keeping pace with the latest generations of HBM.

- **Variability of read/write ratio in inference:** As context windows expand and workloads shift to read-heavy inference, occasional write operations in KV cache remain essential: if no prefix cache hit occurs (i.e., the prompt / prefix is not found in the existing KV cache), the read/write ratio is effectively zero. Although this ratio will likely become increasingly rare over time and inference remains highly read-dominant, the variability of write operation frequency (and the resulting meaningful bandwidth penalty for NAND-based HBF) could render the effective bandwidth of HBF unpredictable in the early phases of any rollout.
- **Endurance, energy intensity, and thermal concerns:** Because it is based on NAND flash, HBF tolerates a finite number of program/erase cycles - whereas DRAM-based HBM offers far greater write endurance. Also, the higher density of a HBF cube result drives higher energy consumption per stack (~160 W per cube in SK Hynix's H^3 architecture, ~4X more power than HBM3E vs. ~2.5X more power than HBM4). Given the peak power of most flagship accelerators of ~1,000-1,200 W, and the utilization of 8 HBF stacks in H^3, the power footprint of HBF alone can easily approach the accelerator's own constraints and result in new thermal issues. This excess heat may further accelerate wear on the flash cells and complicate lifespan considerations.
- **Manufacturing resource intensity:** As with HBM, HBF uses design changes to maximize bandwidth, including increased internal structural parallelism and 3D packaging methods. It also uses ultra-wide on-package interconnect to alleviate the external data bottleneck. These advanced packaging techniques, the use of hybrid bonding, and the ultra-wide interconnect all contribute to a more resource-intensive and higher-cost manufacturing process, including a significantly higher wafer consumption (most estimates suggest a 3X-4X higher cost relative to traditional NAND, similar to the DRAM/HBM "trade ratio").

In aggregate, we view HBF's potential bandwidth limitations and resource intensity as the most significant challenges facing wide adoption of the technology. As context windows lengthen and the share of inference workflows grows, we see improved visibility on the read/write ratio in KV cache and on pricing as key variables.

Alternative approaches to scaling the memory wall - XL Flash and zNAND-O

We are constructive on the broad set of NAND-based technologies set to address memory constraints in AI inference - and see alternative approaches such as XL Flash and zNAND-O ultimately driving similar implications. While HBF remains one of the leading approaches to addressing current AI system memory constraints given public backing from SanDisk and SK Hynix (and ongoing development from Samsung) and significant peer-reviewed publications, other approaches have emerged, including

"XL Flash." Introduced by Kioxia and built on the company's BiCS Flash 3D NAND architecture, the approach includes key architectural and design differences relative to HBF which could potentially offer easier packaging integration and/or lower cost, potentially at the expense of some performance. We would also highlight Samsung's development of zNAND-O - similar in concept such as using TSV to connect 4 or 8 layer vertical stacks of V-NAND dies, while targeting a different market segment in on-device AI (smartphones, PCs, etc.). Samsung is also developing zHBM for data centers where it stacks the memory on top of the AI accelerator rather than next to it, to improve the performance and power efficiency. Although it is too early to tell which approach might come to market first, we believe that all three approaches are ultimately aimed at solving the "memory wall" problem by applying NAND-based system approaches to increase context length while offering significant potential cost savings at the system level.

XL flash may hold write, cost, and certain other advantages over HBF, although based on publicly-available literature HBF appears well-suited for read-heavy AI inference applications. XL Flash utilizes standard PCIe, NVMe, and CXL system buses, but relies on less dense SLC-based NAND, smaller transfer blocks, and larger die architectures relative to traditional 3D NAND flash to maximize bandwidth, significantly lower random read latency, and meaningfully increase write endurance; characteristics that may render XL Flash better suited for write operations relative to HBF. The technology's reliance on existing approaches to 3D NAND manufacturing, its lack of TSVs (through-silicon vias, required for the stacking associated with HBF), and physical separation from the GPU (vs. HBF using the same silicon interposer) may also lower the upfront transition cost and perceived investment risks.

Individual stock implications:

Exhibit 10: Select stocks within our coverage positioned to be affected by HBF

Company	Ticker	Rating	12 month price target	Current Price	Upside/ downside
Advanced Micro Devices	AMD	Buy	$700	$611	14.5%
Broadcom	AVGO	Buy	$540	$363	48.9%
Kioxia	285A.T	Buy	¥38,667	¥17,505	120.9%
Micron	MU	Neutral	$1,250	$1,027	21.7%
Nvidia	NVDA	Buy	$300	$229	30.7%
Samsung	005930.KS	Buy	₩500,000	₩262,000	90.8%
Sandisk	SNDK	Buy	$2,200	$1,607	36.9%
SK Hynix	000660.KS	Buy	₩3,500,000	₩1,681,000	108.2%

Note: Current Price is as of October 9, 2026.

Source: Goldman Sachs Global Investment Research

- **SanDisk (SNDK, Buy):** We remain constructive on SNDK and view the company as a well-positioned beneficiary of the potential widespread adoption of HBF given its technology leadership in collaboration with SK Hynix and its low capital intensity through its Kioxia manufacturing partnership - which we see as particularly differentiating given the resource intensity of HBF. We also believe the company can simultaneously retain its solid core business in traditional NAND, which we expect to continue to outperform on improving product mix and further progress on its BiCS roadmap.

- **Kioxia Holdings (285A.T, Buy):** We reiterate our Buy rating on Kioxia given our expectations on ongoing supply shortages in the NAND market and constructive pricing outlook, which we believe should lead to positive street estimate revisions. We also believe Kioxia is poised to announce a series of shareholder return, including the resumption of dividend payment as well as continued share buyback. Regarding XL-Flash, which is intended to expand the memory capacity of GPUs in inference GPU servers, Kioxia plans to ship samples of a product achieving over 10 million IOPS (storage processing speed) by the end of CY26. It also plans to ship samples of a second-generation product with processing speed further increased to 100 million IOPS in CY27. Compared to HB Flash (HBF), Kioxia's SSD-based solution is differentiated in that it does not require additional manufacturing processes such as TSV.
- **SK Hynix (000660.KS, Buy):** We maintain our positive view on Hynix and see the company as a potential leader in the HBF market. While the company's focus remains on higher-margin DRAM and HBM, Hynix is uniquely positioned to be a prime beneficiary of HBF market expansion given its proactive initiatives in co-development and industry standardization of HBF with SanDisk. Through leveraging its advanced NAND technology, vast experience in producing various types of NAND, and the advanced packaging expertise from HBM leadership, we believe Hynix possesses the core competencies required for HBF commercialization.
- **Samsung Electronics (005930.KS, Buy, on CL):** We maintain our Buy rating (on the APAC Conviction List) on SEC. Although SEC has not yet released specific details on its work on HBF (although it did recently file a trademark application for "zHBF"), recent announcements on its technology milestones—such as the TSV-integrated zNAND-O architecture—demonstrate its active R&D in high-bandwidth NAND. Supported by its largest NAND capacity globally and technology leadership, we believe SEC will likely be ready to compete in the high-bandwidth NAND market, also leveraging its superior cost management and packaging competitiveness. We highlight SEC being the only company in the world that can produce DRAM and NAND while having the foundry/logic/advanced packaging capabilities, which we believe along with the industry's largest memory capacity (for both DRAM and NAND), will help the company to continue to lead across an increasingly multi-tiered AI memory hierarchy.
- **Micron (MU, Neutral):** While we expect the company to participate in any significant future HBF market, we view its focus on HBM and DRAM, its lack of participation in the consortium governing HBF, and its prioritization of DRAM greenfield capacity overall as likely to temper its potential future market share in the novel NAND technology. We are Neutral-rated on MU, primarily on the risks of HBM pricing momentum slowing in 2028 given the prospects of meaningful supply additions by the industry, as well as possible de-specing of HBM as HBF adoption ramps. We nevertheless see continued tailwinds for Micron's business overall, but view SK Hynix and SanDisk as more direct and nearer-term beneficiaries of any HBF-related inflection in NAND demand.
- **Semi Cap Equipment:** We view HBF as a modest incremental positive for semiconductor equipment demand, though the implications are more nuanced than

a simple increase in NAND content. On one hand, HBF introduces additional manufacturing complexity through TSVs, hybrid bonding, advanced packaging, inspection, and test, which could benefit equipment suppliers with exposure to packaging and integration workflows. On the other hand, HBF's value proposition is rooted in reducing memory capacity bottlenecks and improving GPU utilization, which could ultimately lower the number of accelerators required for a given inference workload. We nevertheless see the added packaging, integration, and memory intensity as likely to offset any reduction in system-level hardware demand.

- **Digital Semis:** We see an incrementally negative impact for a range of XPU accelerator makers, including Nvidia, Broadcom, AMD, and others - as the success of any high-performance NAND approach is likely to reduce the number of accelerators ultimately deployed at a similar performance level given the potential cost savings involved. We would point out that Nvidia is currently the only provider of accelerators which controls the entire system stack from software to hardware components - and hence it may be able to optimize the system-level design to mitigate this impact, similar to what it has done by optimizing the HBM-NAND balance in the current Rubin generation.

Disclosure Appendix

Reg AC

We, James Schneider, Ph.D., Khalil Fenina, Giuni Lee, Shuhei Nakamura, Allen Chang, Daiki Takayama, Marshall Wong, Anmol Makkar, Luya You and Liana Wadhwani, hereby certify that all of the views expressed in this report accurately reflect our personal views about the subject company or companies and its or their securities. We also certify that no part of our compensation was, is or will be, directly or indirectly, related to the specific recommendations or views expressed in this report.

Unless otherwise stated, the individuals listed on the cover page of this report are analysts in Goldman Sachs' Global Investment Research division.

Contributing Authors: James Schneider, Ph.D. Goldman Sachs & Co. LLC, Khalil Fenina Goldman Sachs & Co. LLC, Giuni Lee Goldman Sachs (Singapore) Pte, Shuhei Nakamura Goldman Sachs Japan Co., Ltd., Allen Chang Goldman Sachs (Asia) L.L.C., Daiki Takayama Goldman Sachs Japan Co., Ltd., Marshall Wong Goldman Sachs & Co. LLC, Anmol Makkar Goldman Sachs & Co. LLC, Luya You Goldman Sachs & Co. LLC, Liana Wadhwani Goldman Sachs & Co. LLC.

Unless otherwise stated, the individuals listed in the Contributing Authors disclosure of this report are analysts in Goldman Sachs' Global Investment Research division.

GS Factor Profile

The Goldman Sachs Factor Profile provides investment context for a stock by comparing key attributes to the market (i.e. our universe of rated stocks) and its sector peers. The four key attributes depicted are: Growth, Financial Returns, Multiple (e.g. valuation) and Integrated (a composite of Growth, Financial Returns and Multiple). Growth, Financial Returns and Multiple are calculated by using normalized ranks for specific metrics for each stock. The normalized ranks for the metrics are then averaged and converted into percentiles for the relevant attribute. The precise calculation of each metric may vary depending on the fiscal year, industry and region, but the standard approach is as follows:

Growth is based on a stock's forward-looking sales growth, EBITDA growth and EPS growth (for financial stocks, only EPS and sales growth), with a higher percentile indicating a higher growth company. **Financial Returns** is based on a stock's forward-looking ROE, ROCE and CROCI (for financial stocks, only ROE), with a higher percentile indicating a company with higher financial returns. **Multiple** is based on a stock's forward-looking P/E, P/B, price/dividend (P/D), EV/EBITDA, EV/FCF and EV/Debt Adjusted Cash Flow (DACF) (for financial stocks, only P/E, P/B and P/D), with a higher percentile indicating a stock trading at a higher multiple. The **Integrated** percentile is calculated as the average of the Growth percentile, Financial Returns percentile and (100% - Multiple percentile).

Financial Returns and Multiple use the Goldman Sachs analyst forecasts at the fiscal year-end at least three quarters in the future. Growth uses inputs for the fiscal year at least seven quarters in the future compared with the year at least three quarters in the future (on a per-share basis for all metrics).

For a more detailed description of how we calculate the GS Factor Profile, please contact your GS representative.

M&A Rank

Across our global coverage, we examine stocks using an M&A framework, considering both qualitative factors and quantitative factors (which may vary across sectors and regions) to incorporate the potential that certain companies could be acquired. We then assign a M&A rank as a means of scoring companies under our rated coverage from 1 to 3, with 1 representing high (30%-50%) probability of the company becoming an acquisition target, 2 representing medium (15%-30%) probability and 3 representing low (0%-15%) probability. For companies ranked 1 or 2, in line with our standard departmental guidelines we incorporate an M&A component into our target price. M&A rank of 3 is considered immaterial and therefore does not factor into our price target, and may or may not be discussed in research.

Quantum

Quantum is Goldman Sachs' proprietary database providing access to detailed financial statement histories, forecasts and ratios. It can be used for in-depth analysis of a single company, or to make comparisons between companies in different sectors and markets.

Disclosures

Distribution of ratings/investment banking relationships

Goldman Sachs Investment Research global Equity coverage universe

	Rating Distribution			Investment Banking Relationships		
	Buy	Hold	Sell	Buy	Hold	Sell
Global	50%	34%	16%	65%	59%	43%

As of July 1, 2026, Goldman Sachs Global Investment Research had investment ratings on 3,104 equity securities. Goldman Sachs assigns stocks as Buys and Sells on various regional Investment Lists; stocks not so assigned are deemed Neutral. Such assignments equate to Buy, Hold and Sell for the purposes of the above disclosure required by the FINRA Rules. See 'Ratings, Coverage universe and related definitions' below. The Investment Banking Relationships chart reflects the percentage of subject companies within each rating category for whom Goldman Sachs has provided investment banking services within the previous twelve months.

Regulatory disclosures

Disclosures required by United States laws and regulations

See company-specific regulatory disclosures above for any of the following disclosures required as to companies referred to in this report: manager or co-manager in a pending transaction; 1% or other ownership; compensation for certain services; types of client relationships; managed/co-managed public offerings in prior periods; directorships; registered market maker. Goldman Sachs trades or may trade as a principal in debt securities (or in related derivatives) of issuers discussed in this report.

The following are additional required disclosures: **Ownership and material conflicts of interest:** Goldman Sachs policy prohibits its analysts, professionals reporting to analysts and members of their households from owning securities of any company in the analyst's area of coverage. **Analyst compensation:** Analysts are paid in part based on the profitability of Goldman Sachs, which includes investment banking revenues. **Analyst as officer or director:** Goldman Sachs policy generally prohibits its analysts, persons reporting to analysts or members of their households from serving as an officer, director or advisor of any company in the analyst's area of coverage. **Non-U.S. Analysts:** Non-U.S. analysts may not be associated persons of

Goldman Sachs & Co. LLC and therefore may not be subject to FINRA Rule 2241 or FINRA Rule 2242 restrictions on communications with a subject company, public appearances and trading in securities covered by the analysts.

Additional disclosures required under the laws and regulations of jurisdictions other than the United States

The following disclosures are those required by the jurisdiction indicated, except to the extent already made above pursuant to United States laws and regulations. **Australia:** Goldman Sachs Australia Pty Ltd and its affiliates are not authorised deposit-taking institutions (as that term is defined in the Banking Act 1959 (Cth)) in Australia and do not provide banking services, nor carry on a banking business, in Australia. This research, and any access to it, is intended only for "wholesale clients" within the meaning of the Australian Corporations Act, unless otherwise agreed by Goldman Sachs. In producing research reports, members of Global Investment Research of Goldman Sachs Australia may attend site visits and other meetings hosted by the companies and other entities which are the subject of its research reports. In some instances the costs of such site visits or meetings may be met in part or in whole by the issuers concerned if Goldman Sachs Australia considers it is appropriate and reasonable in the specific circumstances relating to the site visit or meeting. To the extent that the contents of this document contains any financial product advice, it is general advice only and has been prepared by Goldman Sachs without taking into account a client's objectives, financial situation or needs. A client should, before acting on any such advice, consider the appropriateness of the advice having regard to the client's own objectives, financial situation and needs. A copy of certain Goldman Sachs Australia and New Zealand disclosure of interests and a copy of Goldman Sachs' Australian Sell-Side Research Independence Policy Statement are available at: https://www.goldmansachs.com/disclosures/australia-new-zealand/index.html. **Brazil:** Disclosure information in relation to CVM Resolution n. 20 is available at https://www.gs.com/worldwide/brazil/area/gir/index.html. Where applicable, the Brazil-registered analyst primarily responsible for the content of this research report, as defined in Article 20 of CVM Resolution n. 20, is the first author named at the beginning of this report, unless indicated otherwise at the end of the text. **Canada:** This information is being provided to you for information purposes only and is not, and under no circumstances should be construed as, an advertisement, offering or solicitation by Goldman Sachs & Co. LLC for purchasers of securities in Canada to trade in any Canadian security. Goldman Sachs & Co. LLC is not registered as a dealer in any jurisdiction in Canada under applicable Canadian securities laws and generally is not permitted to trade in Canadian securities and may be prohibited from selling certain securities and products in certain jurisdictions in Canada. If you wish to trade in any Canadian securities or other products in Canada please contact Goldman Sachs Canada Inc., an affiliate of The Goldman Sachs Group Inc., or another registered Canadian dealer. **Hong Kong:** Further information on the securities of covered companies referred to in this research may be obtained on request from Goldman Sachs (Asia) L.L.C. **India:** Further information on the subject company or companies referred to in this research may be obtained from Goldman Sachs (India) Securities Private Limited, Research Analyst - SEBI Registration Number INH000001493, 10th Floor, Ascent-Worli, Sudam Kalu Ahire Marg, Worli, Mumbai-400 025, India, Corporate Identity Number U74140MH2006FTC160634, Phone +91 22 6616 9000, Fax +91 22 6616 9001. Goldman Sachs may beneficially own 1% or more of the securities (as such term is defined in clause 2 (h) the Indian Securities Contracts (Regulation) Act, 1956) of the subject company or companies referred to in this research report. Registration granted by SEBI and certification from NISM in no way guarantee performance of the intermediary or provide any assurance of returns to investors. Goldman Sachs (India) Securities Private Limited compliance officer and investor grievance contact details, a copy of the annual compliance audit report and other relevant information and disclosures can be found at this link: https://www.goldmansachs.com/worldwide/india/research-analyst. **Japan:** See below. **Korea:** This research, and any access to it, is intended only for "professional investors" within the meaning of the Financial Services and Capital Markets Act, unless otherwise agreed by Goldman Sachs. Further information on the subject company or companies referred to in this research may be obtained from Goldman Sachs (Asia) L.L.C., Seoul Branch. **New Zealand:** Goldman Sachs New Zealand Limited and its affiliates are neither "registered banks" nor "deposit takers" (as defined in the Reserve Bank of New Zealand Act 1989) in New Zealand. This research, and any access to it, is intended for "wholesale clients" (as defined in the Financial Advisers Act 2008) unless otherwise agreed by Goldman Sachs. A copy of certain Goldman Sachs Australia and New Zealand disclosure of interests is available at: https://www.goldmansachs.com/disclosures/australia-new-zealand/index.html. **Russia:** Research reports distributed in the Russian Federation are not advertising as defined in the Russian legislation, but are information and analysis not having product promotion as their main purpose and do not provide appraisal within the meaning of the Russian legislation on appraisal activity. Research reports do not constitute a personalized investment recommendation as defined in Russian laws and regulations, are not addressed to a specific client, and are prepared without analyzing the financial circumstances, investment profiles or risk profiles of clients. Goldman Sachs assumes no responsibility for any investment decisions that may be taken by a client or any other person based on this research report. **Singapore:** Goldman Sachs (Singapore) Pte. (Company Number: 198602165W), which is regulated by the Monetary Authority of Singapore, accepts legal responsibility for this research, and should be contacted with respect to any matters arising from, or in connection with, this research. **Taiwan:** This material is for reference only and must not be reprinted without permission. Investors should carefully consider their own investment risk. Investment results are the responsibility of the individual investor. **United Kingdom:** Persons who would be categorized as retail clients in the United Kingdom, as such term is defined in the rules of the Financial Conduct Authority, should read this research in conjunction with prior Goldman Sachs research on the covered companies referred to herein and should refer to the risk warnings that have been sent to them by Goldman Sachs International. A copy of these risks warnings, and a glossary of certain financial terms used in this report, are available from Goldman Sachs International on request.

European Union and United Kingdom: Disclosure information in relation to Article 6 (2) of the European Commission Delegated Regulation (EU) (2016/958) supplementing Regulation (EU) No 596/2014 of the European Parliament and of the Council (including as that Delegated Regulation is implemented into United Kingdom domestic law and regulation following the United Kingdom's departure from the European Union and the European Economic Area) with regard to regulatory technical standards for the technical arrangements for objective presentation of investment recommendations or other information recommending or suggesting an investment strategy and for disclosure of particular interests or indications of conflicts of interest is available at https://www.gs.com/disclosures/europeanpolicy.html which states the European Policy for Managing Conflicts of Interest in Connection with Investment Research.

Japan: Goldman Sachs Japan Co., Ltd. is a Financial Instrument Dealer registered with the Kanto Financial Bureau under registration number Kinsho 69, and a member of Japan Securities Dealers Association, Financial Futures Association of Japan Type II Financial Instruments Firms Association, and Investment Management Association of Japan. Sales and purchase of equities are subject to commission pre-determined with clients plus consumption tax. See company-specific disclosures as to any applicable disclosures required by Japanese stock exchanges, the Japanese Securities Dealers Association or the Japanese Securities Finance Company.

Ratings, coverage universe and related definitions

Buy (B), Neutral (N), Sell (S) Analysts recommend stocks as Buys or Sells for inclusion on various regional Investment Lists. Being assigned a Buy or Sell on an Investment List is determined by a stock's total return potential relative to its coverage universe. Any stock not assigned as a Buy or a Sell on an Investment List with an active rating (i.e., a stock that is not Rating Suspended, Not Rated, Early-Stage Biotech, Coverage Suspended or Not Covered), is deemed Neutral. Each region manages Regional Conviction Lists, which are selected from Buy rated stocks on the respective region's Investment Lists and represent investment recommendations focused on the size of the total return potential and/or the likelihood of the realization of the return across their respective areas of coverage. The addition or removal of stocks from such Conviction Lists are managed by the Investment Review Committee or other designated committee in each respective region and do not represent a change in the analysts' investment rating for such stocks.

Total return potential represents the upside or downside differential between the current share price and the price target, including all paid or anticipated dividends, expected during the time horizon associated with the price target. Price targets are required for all covered stocks. The total return potential, price target and associated time horizon are stated in each report adding or reiterating an Investment List membership.

Coverage Universe: A list of all stocks in each coverage universe is available by primary analyst, stock and coverage universe at

https://www.gs.com/research/hedge.html.

Not Rated (NR). The investment rating, target price and earnings estimates (where relevant) are removed pursuant to Goldman Sachs policy when Goldman Sachs is acting in an advisory capacity in a merger or in a strategic transaction involving this company, when there are legal, regulatory or policy constraints due to Goldman Sachs' involvement in a transaction, and in certain other circumstances. **Early-Stage Biotech (ES).** An investment rating and a target price are not assigned pursuant to Goldman Sachs policy when this company has neither a drug, treatment or medical device that has passed a Phase II clinical trial nor a license to distribute a post-Phase II drug, treatment or medical device. **Rating Suspended (RS).** Goldman Sachs Research has suspended the investment rating and price target for this stock, because there is not a sufficient fundamental basis for determining an investment rating or target price. The previous investment rating and target price, if any, are no longer in effect for this stock and should not be relied upon. **Coverage Suspended (CS).** Goldman Sachs has suspended coverage of this company. **Not Covered (NC).** Goldman Sachs does not cover this company.

Global product; distributing entities

Goldman Sachs Global Investment Research produces and distributes research products for clients of Goldman Sachs on a global basis. Analysts based in Goldman Sachs offices around the world produce research on industries and companies, and research on macroeconomics, currencies, commodities and portfolio strategy. This research is disseminated in Australia by Goldman Sachs Australia Pty Ltd (ABN 21 006 797 897); in Brazil by Goldman Sachs do Brasil Corretora de Títulos e Valores Mobiliários S.A.; Public Communication Channel Goldman Sachs Brazil: 0800 727 5764 and / or contatogoldmanbrasil@gs.com. Available Weekdays (except holidays), from 9am to 6pm. Canal de Comunicação com o Público Goldman Sachs Brasil: 0800 727 5764 e/ou contatogoldmanbrasil@gs.com. Horário de funcionamento: segunda-feira à sexta-feira (exceto feriados), das 9h às 18h; in Canada by Goldman Sachs & Co. LLC; in Hong Kong by Goldman Sachs (Asia) L.L.C.; in India by Goldman Sachs (India) Securities Private Ltd.; in Japan by Goldman Sachs Japan Co., Ltd.; in the Republic of Korea by Goldman Sachs (Asia) L.L.C., Seoul Branch; in New Zealand by Goldman Sachs New Zealand Limited; in Russia by OOO Goldman Sachs; in Singapore by Goldman Sachs (Singapore) Pte. (Company Number: 198602165W); and in the United States of America by Goldman Sachs & Co. LLC. Goldman Sachs International has approved this research in connection with its distribution in the United Kingdom.

Goldman Sachs International ("GSI"), authorised by the Prudential Regulation Authority ("PRA") and regulated by the Financial Conduct Authority ("FCA") and the PRA, has approved this research in connection with its distribution in the United Kingdom.

European Economic Area: Goldman Sachs Bank Europe SE ("GSBE") is a credit institution incorporated in Germany and, within the Single Supervisory Mechanism, subject to direct prudential supervision by the European Central Bank and in other respects supervised by German Federal Financial Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, BaFin) and Deutsche Bundesbank and disseminates research within the European Economic Area.

General disclosures

This research is for our clients only. Other than disclosures relating to Goldman Sachs, this research is based on current public information that we consider reliable, but we do not represent it is accurate or complete, and it should not be relied on as such. The information, opinions, estimates and forecasts contained herein are as of the date hereof and are subject to change without prior notification. We seek to update our research as appropriate, but various regulations may prevent us from doing so. Other than certain industry reports published on a periodic basis, the large majority of reports are published at irregular intervals as appropriate in the analyst's judgment.

Goldman Sachs conducts a global full-service, integrated investment banking, investment management, and brokerage business. We have investment banking and other business relationships with a substantial percentage of the companies covered by Global Investment Research. Goldman Sachs & Co. LLC, the United States broker dealer, is a member of SIPC (https://www.sipc.org).

Our salespeople, traders, and other professionals may provide oral or written market commentary or trading strategies to our clients and principal trading desks that reflect opinions that are contrary to the opinions expressed in this research. Our asset management area, principal trading desks and investing businesses may make investment decisions that are inconsistent with the recommendations or views expressed in this research.

The analysts named in this report may have from time to time discussed with our clients, including Goldman Sachs salespersons and traders, or may discuss in this report, trading strategies that reference catalysts or events that may have a near-term impact on the market price of the equity securities discussed in this report, which impact may be directionally counter to the analyst's published price target expectations for such stocks. Any such trading strategies are distinct from and do not affect the analyst's fundamental equity rating for such stocks, which rating reflects a stock's return potential relative to its coverage universe as described herein.

We and our affiliates, officers, directors, and employees will from time to time have long or short positions in, act as principal in, make a market or act as liquidity provider in, and buy or sell, any securities, derivatives, or other financial instruments referred to in this research (including any financial instruments issued by the companies referred to herein), unless otherwise prohibited by regulation or Goldman Sachs policy.

The views attributed to third party presenters at Goldman Sachs arranged conferences, including individuals from other parts of Goldman Sachs, do not necessarily reflect those of Global Investment Research and are not an official view of Goldman Sachs.

Any third party referenced herein, including any salespeople, traders and other professionals or members of their household, may have positions in the products mentioned that are inconsistent with the views expressed by analysts named in this report.

This research is not an offer to sell or the solicitation of an offer to buy any security in any jurisdiction where such an offer or solicitation would be illegal. It does not constitute a personal recommendation or take into account the particular investment objectives, financial situations, or needs of individual clients. Clients should consider whether any advice or recommendation in this research is suitable for their particular circumstances and, if appropriate, seek professional advice, including tax advice. The price and value of investments referred to in this research and the income from them may fluctuate. Past performance is not a guide to future performance, future returns are not guaranteed, and a loss of original capital may occur. Fluctuations in exchange rates could have adverse effects on the value or price of, or income derived from, certain investments.

Certain transactions, including those involving futures, options, and other derivatives, give rise to substantial risk and are not suitable for all investors. Investors should review current options and futures disclosure documents which are available from Goldman Sachs sales representatives or at https://www.theocc.com/about/publications/character-risks.jsp and https://www.goldmansachs.com/disclosures/cftc_fcm_disclosures. Transaction costs may be significant in option strategies calling for multiple purchase and sales of options such as spreads. Supporting documentation will be supplied upon request.

Differing Levels of Service provided by Global Investment Research: The level and types of services provided to you by Goldman Sachs Global Investment Research may vary as compared to that provided to internal and other external clients of GS, depending on various factors including your individual preferences as to the frequency and manner of receiving communication, your risk profile and investment focus and perspective (e.g., marketwide, sector specific, long term, short term), the size and scope of your overall client relationship with GS, and legal and regulatory constraints. As an example, certain clients may request to receive notifications when research on specific securities is published, and certain clients may request that specific data underlying analysts' fundamental analysis available on our internal client websites be delivered to them electronically through data

feeds or otherwise. No change to an analyst's fundamental research views (e.g., ratings, price targets, or material changes to earnings estimates for equity securities), will be communicated to any client prior to inclusion of such information in a research report broadly disseminated through electronic publication to our internal client websites or through other means, as necessary, to all clients who are entitled to receive such reports.

All research reports are disseminated and available to all clients simultaneously through electronic publication to our internal client websites. Not all research content is redistributed to our clients or available to third-party aggregators, nor is Goldman Sachs responsible for the redistribution of our research by third party aggregators. For research, models or other data related to one or more securities, markets or asset classes (including related services) that may be available to you, please contact your GS representative or go to https://research.gs.com.

Disclosure information is also available at https://www.gs.com/research/hedge.html or from Research Compliance, 200 West Street, New York, NY 10282.

© 2026 Goldman Sachs.

You are permitted to store, display, analyze, modify, reformat, and print the information made available to you via this service only for your own use. You may not resell or reverse engineer this information to calculate or develop any index for disclosure and/or marketing or create any other derivative works or commercial product(s), data or offering(s) without the express written consent of Goldman Sachs. You are not permitted to publish, transmit, or otherwise reproduce this information, in whole or in part, in any format to any third party without the express written consent of Goldman Sachs. This foregoing restriction includes, without limitation, using, extracting, downloading or retrieving this information, in whole or in part, to train or finetune a machine learning or artificial intelligence system, or to provide or reproduce this information, in whole or in part, as a prompt or input to any such system.